

L8b: Option Contracts: Call and Put Payoff and Profit

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Objectives for Today

Today opens Unit 3. Units 1 and 2 modeled assets we hold directly; a **derivative** is instead a contract whose cash flows are a function of someone else's price. We read the contract, compute what it pays at expiration, and keep what the contract promises separate from what the investor earns.

Today's concepts

- **Read an option contract**: underlying, strike K , expiration T , exercise style, deliverable q , direction θ , and the premium: which are contractual, which is quoted.
- **Separate payoff from profit**: the expiration payoff of all four positions, the premium adjustment and the convention it hides, the breakeven prices, and the maximum profit and loss of each.
- **Compose positions**: a strategy as a signed sum of legs, with a corner at each strike where the legs do not cancel and an offset set by the net debit or credit.

Lectures and Examples

Lecture:

CHEME-5660-L8b-Lecture-IntroductionToDerivativesContracts-Fall-2026.ipynb

Example 1:

CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:

CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

The first example builds all four single-contract positions from a real options chain, sweeps the full domain $S_T \geq 0$, converts per-share quantities to per-contract dollars, and verifies every breakeven and bound numerically; the second combines signed legs into vertical spreads, straddles, and strangles.

Concept Review: From Assets to Contracts

Written on Assets

Units 1 and 2 priced and allocated assets we hold **directly**: Treasury cash flows (L2a, L2b), equity growth rates (L3a, L4b), and portfolios built from them (L5b through L7b).

Carried forward: the price of the underlying. Let $S_t \geq 0$ be that price in USD/share at time t , so S_0 is today and S_T is expiration.

- **The payoff is contractual, not statistical:** given S_T , what the contract pays is a formula written into it. No estimation enters that step. Uncertainty enters only through S_T .
- **The two sides are not symmetric:** one counterparty holds a right, the other carries an obligation. The premium is the price of that asymmetry, and it is paid whether or not the right is ever used.

What Is a Derivative?

A **derivative** is a contract whose cash flows depend on an underlying quantity: an equity price, a rate, a currency, a commodity, an index, or a credit event. The families differ in who is obligated:

- **Futures and forwards:** *both* sides must transact the underlying later at a set price. Futures are exchange-traded and standardized; forwards are negotiated over the counter.
- **Swaps:** *both* sides must exchange two streams of cash flows. Traded over the counter.
- **Options:** the buyer holds a *right*, and only the seller carries an obligation. Listed equity options are exchange-traded and centrally cleared; an over-the-counter market also exists.

That asymmetry is the subject of the lecture: it makes an option payoff a **kinked** function instead of a straight line, and it is what the premium buys.

Reading an Option Contract

Seven quantities must be pinned down before any payoff can be computed:

- the **underlying** asset
- $K > 0$: strike, USD/share
- $T > 0$: expiration, years
- **exercise style**: European or American
- $q \in \mathbb{N}$: deliverable, shares/contract
- $\theta \in \{-1, +1\}$: direction, +1 long, -1 short
- $C_0, P_0 \geq 0$: call and put premium, USD/share

Only the **premium** is a market quantity: it is quoted and it moves, and L9a is where it comes from. The rest are contractual or chosen when you trade.

- **Exercise style**: European exercises only at T , American on or before T . At expiration the two behave identically, so it does not enter today; it affects the premium (L9a, L9b).
- **The deliverable q** : standard contracts use $q = 100$, but the specification controls. A premium quoted at 21.58 USD/share is 2,158 USD of cash for one contract.

Reading an Option Contract: The Cash Flows

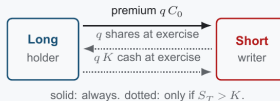
An option contract: a right and an obligation

The premium is paid once, at the open, whether or not the right is ever used

Call contract

THE RIGHT TO BUY AT K

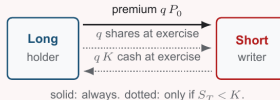
- **Long (holder)** pays C_0 /share for the right to buy at K .
- **Short (writer)** collects C_0 , must deliver at K if assigned.
- At T , exercised when $S_T > K$: payoff $(S_T - K)^+$.



Put contract

THE RIGHT TO SELL AT K

- **Long (holder)** pays P_0 /share for the right to sell at K .
- **Short (writer)** collects P_0 , must buy at K if assigned.
- At T , exercised when $S_T < K$: payoff $(K - S_T)^+$.



European style exercises only at T ; American style on or before T . Commonly $q = 100$ shares/contract.

One flow is unconditional, the premium at the open; two are conditional on exercise. Long positions risk only the premium, short ones collect it and carry the obligation.

Expiration Payoff Is Contractual

For any real x write $x^+ \equiv \max(x, 0)$. With $S_T \geq 0$ the underlying price at expiration and $K > 0$ the strike, both in USD/share, the payoff **per share** to the long holder is given by:

$$h_c(S_T; K) = (S_T - K)^+, \quad h_p(S_T; K) = (K - S_T)^+$$

The call holder exercises only when $S_T > K$, the put holder only when $S_T < K$, and otherwise abandons the right. Because the holder may always decline, **both payoffs are nonnegative**: the signature of a right.

- **The short side mirrors it**: with direction $\theta \in \{-1, +1\}$, the signed payoff is θh_c or θh_p .
- **Moneyness** classifies exercise value, not profit: a call is in the money when $S_T > K$, out of the money when $S_T < K$, and the put reverses these. It says nothing about whether the trade made money.

Profit Requires a Cash-Flow Convention

Payoff belongs to the contract. Profit belongs to the investor, and it needs a rule for the premium, which moved at a different time. Under the **undiscounted convention**, with $C_0, P_0 \geq 0$ the premiums in USD/share and θ the direction, profit per share is given by:

$$\Pi_T^{c,\theta}(S_T) = \theta \left(\underbrace{h_c(S_T; K)}_{\text{payoff at } T} - \underbrace{C_0}_{\text{premium at 0}} \right), \quad \Pi_T^{p,\theta}(S_T) = \theta \left(h_p(S_T; K) - P_0 \right)$$

For a long position the premium is an outflow, for a short an inflow: one θ multiplies both terms.

- **Say what it hides**: this mixes time-0 and time- T dollars, which every earlier lecture avoided. A time- T P&L would grow the premium forward, a time-0 value would discount the payoff (L1b).
- **Why use it**: it gets the shape and the breakeven right with the fewest moving parts. A convention, not an accounting identity.

Profit: Where a Position Breaks Even

The **breakeven** S_{BE} , in USD/share, is the terminal price at which profit is exactly zero. Setting $\Pi_T = 0$ on the exercised branch gives $S_T - K - C_0 = 0$ for the long call and $K - S_T - P_0 = 0$ for the long put, hence:

$$S_{BE}^c = K + C_0, \quad S_{BE}^p = K - P_0$$

- **The breakeven is not the strike**: the premium shifts the curve down (long) or up (short) without moving the kink at K . On the AMD chain a $K = 230$ call at $C_0 = 21.58$ breaks even at 251.58: finishing in the money is not the same as making money.
- **Both directions share it**: short profit is the exact negative of long profit, so both sides agree on the breakeven price.

Example 1: CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Loss Bounds and Direction Matter

All per share; multiply by q and by the number of contracts for account dollars.

Position	Maximum profit	Maximum loss	Breakeven
Long call	unbounded as $S_T \rightarrow \infty$	C_0	$K + C_0$
Short call	C_0	unbounded as $S_T \rightarrow \infty$	$K + C_0$
Long put	$K - P_0$ at $S_T = 0$	P_0	$K - P_0$
Short put	P_0	$K - P_0$ at $S_T = 0$	$K - P_0$

The difference between the call rows and the put rows is the **domain**: $S_T \geq 0$, so $h_p \leq K$ is capped by the strike, while h_c has no cap because S_T has none.

- **Only one loss is genuinely unlimited**: the uncovered short call. The long call's profit is unbounded too, but no loss is.
- **A short put's loss is finite**: $K - P_0$ at $S_T = 0$. On the $K = 220$ contract collecting 16.33, that is 203.67 USD/share, or 20,367 USD per contract. **Large is not unlimited**.

Composite Positions Add Leg by Leg

Let position $\mathcal{C}$ contain legs $i = 1, \dots, d$ on one underlying and one expiration. Leg i has strike $K_i > 0$, direction $\theta_i \in \{-1, +1\}$, copies $n_i \in \mathbb{N}$, premium $P_i \geq 0$ in USD/share, and long per-share payoff $h_i(S_T)$. The **per-share** totals are given by:

$$H_{\mathcal{C}}(S_T) = \sum_{i=1}^d \underbrace{\theta_i n_i}_{\text{signed copies of leg } i} h_i(S_T), \quad \Pi_{\mathcal{C}}(S_T) = \sum_{i=1}^d \theta_i n_i [h_i(S_T) - P_i]$$

Both are in USD/share. For account dollars, multiply leg i by its deliverable $q_i \in \mathbb{N}$ shares/contract.

- **The kinks are the strikes:** each h_i bends only at K_i , so the profit is piecewise linear, with a corner at strike K exactly when the aggregate jump $\sum_{i:K_i=K} \theta_i n_i$ is nonzero. Opposing legs at one strike cancel.
- **The offset is the net premium:** $\sum_i \theta_i n_i P_i$, the net debit or credit, shifts the diagram without changing its shape.

Legs must share an underlying and an expiration for one diagram in S_T .

Composite Positions: Three Structures, One Sum

Three structures, D the total premium per share:

- **Vertical spread**: two legs of one type, different strikes, opposite directions. Both extremes finite.
- **Straddle**: a call and a put at one strike K , same direction. Profit $\theta(|S_T - K| - D)$; breakevens $K + D$, and $K - D$ when $D \leq K$.
- **Strangle**: a call and a put at strikes $K_1 < K_2$, same direction. A flat plateau replaces the straddle's single kink.

State the bounds correctly. A long straddle is often said to have “unlimited profit in either direction.” **That is wrong on the downside.** Since $S_T \geq 0$ the put leg cannot pay more than its strike: the AMD $K = 230$ straddle with $D = 42.88$ caps its downside at 187.12 USD/share.

Example 2: CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

Optional Advanced Material

Advanced: [advanced/static_replication/](#)

[CHEME-5660-L8b-Advanced-Static-Replication-Fall-2026.ipynb](#)

Optional, not a prerequisite for L9a. The composite rule says adding legs **produces** payoff shapes; the notebook proves the converse. Every continuous piecewise-linear payoff with kinks $0 < K_1 < \dots < K_d$ is $a + b S_T + \sum_i w_i (S_T - K_i)^+$ in **exactly one** way, with a the payoff at zero, b the initial slope, and w_i the jump in slope at K_i . It proves existence by a telescoping sum of slopes and uniqueness by sweeping out from the leftmost interval, derives the put payoff identity, tabulates every structure above as one triple (a, b, w) , and marks the boundary of what a single expiration can reach.

Notebook: [lectures/week-8/L8b/advanced](#), also linked from the lecture notebook.

Summary

Today we translated contract terms into expiration cash flows and kept the contractual payoff separate from the investor's premium-dependent profit.

Key takeaways

- **Payoff belongs to the contract:** a long call pays $(S_T - K)^+$ and a long put pays $(K - S_T)^+$ per share, and short positions reverse those flows exactly. Only S_T is uncertain.
- **Profit needs a convention:** premium, financing time, deliverable q , fees, and direction θ all enter, so strike, moneyness, and breakeven are three different things.
- **Loss bounds are asymmetric, and only one side is unlimited:** the uncovered short call has no finite bound because S_T has none; a standard short put's worst case is finite at $K - P_0$, reached at $S_T = 0$.

Next, L9a values European claims with Black–Scholes–Merton and American claims with a Cox–Ross–Rubinstein lattice: where the premium comes from.

Today: what the contract promises, what the investor earns,
and why only one of the four positions can lose without
bound.